

KSHITIJ DURAPHE

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Machine learning engineer working across the full model stack — custom **CUDA kernels** and GPU optimization, **distributed training**, **RL post-training**, and edge inference — with a parallel research record in **mechanistic interpretability** and **foundation-model evaluation**. First author on three 2026 interpretability papers; co-first author on a [NeurIPS ML4PS 2025 Spotlight](#) (top 1%); first author in *The Astrophysical Journal*. M.S. Boston University.

PROFESSIONAL EXPERIENCE

Position Imaging

Stratham, NH

Machine Learning Engineer

May 2026 – Present

- Developed and productionized **graph neural networks** for RFID localization in multipath-dominated environments, owning the end-to-end stack: architecture, training, GPU optimization, edge deployment, and production monitoring.
- Wrote custom **CUDA kernels** and GPU-level optimizations for performance-critical paths in GNN training and inference.
- Built the team's internal **ML experimentation platform** (ClearML, Azure, Datadog) for reproducible training and evaluation, sustaining **30+ experiments/day**.
- Designed training systems using **reinforcement learning** and **neural surrogate models** to cut the cost of computationally expensive GNN workloads, making large sweeps over architectures and training strategies affordable.
- Built edge inference pipelines end to end — model compilation, graph/operator optimization, hardware-specific acceleration, and integration with custom inference runtimes.
- Developed production **ML observability** infrastructure for model behaviour, inference performance, and system health, giving product and support teams tooling to diagnose deployed models.

Thespian Labs

Somerville, MA

AI Engineer — Foundation Models and RL Post-Training

Nov 2025 – Feb 2026

- Trained **Text2Motion** foundation models with **VQ-VAE** tokenization; ran multimodal pre-training over 5000+ hours of human-performance time-series on GCP.
- Implemented DARTControl-style **reinforcement-learning post-training** for controllable motion synthesis, including reward modeling and rollout infrastructure.
- Released the open-source MLOps library **jobber** for programmatic research-job submission to cloud GPU platforms.

Absentia Technologies

Boston, MA

Founding Machine Learning Engineer

Jan 2025 – Nov 2025

- Implemented **distributed training** with PyTorch **FSDP** (multi-GPU, mixed precision, sharding, checkpointing, fault recovery) to train vision-language models beyond single-device memory.
- Established an automated **evaluation pipeline** benchmarking model accuracy and API latency; reduced regressions **40%** and raised production stability to **99.9%**, including diagnosing a production memory leak on call.
- Architected the production platform on AWS from scratch with Terraform and Docker CI/CD, cutting deployment cycles from days to **<2 hours**; built a diffusion-based synthetic-data pipeline that reduced false positives **15%**.

The KeelWorks Foundation

Oak Harbor, WA (Remote)

Software Engineer, ML Applications

Jul 2024 – Jan 2025

- Reduced model size **50%** and improved inference speed **80%** via **8-bit GPTQ quantization** and knowledge distillation while holding **>90%** task accuracy; fine-tuned **Mistral-7B** for synthetic-data generation.

Halo AI (Columbia-incubated stealth)

New York, NY (Remote)

Founding AI Engineer

Dec 2023 – Aug 2024

- Researched on-device **federated** and **ensemble** LLMs under tight compute and memory budgets; systematic study of pruning, distillation, and post-training quantization trade-offs across latency, footprint, and accuracy.

RESEARCH

UniverseTBD Collaboration

Remote

Researcher — Interpretability and Foundation-Model Evaluation

Jul 2024 – Present

- Co-first author on *The Platonic Universe* ([Spotlight](#), [top 1%](#) at NeurIPS ML4PS 2025; expanded version submitted to ICLR 2027). Designed the representational-alignment **evaluation framework** over eleven model families from 10M to 10B parameters across JWST,

HSC, Legacy Survey imaging and DESI spectroscopy; showed physics performance tracks **local** embedding geometry (mutual k -NN) but not **global** similarity (CKA).

- First author on two further interpretability papers (Sci-FM @ COLM 2026; NeurIPS Interpretability for Discovery 2026), establishing that model capabilities emerge in a fixed order tracking their physical difficulty, and separating layerwise where representations agree geometrically from where they agree physically.
- Creator and maintainer of **platonic-universe**: survey-aware loaders, reproducible eval harness, configs, CI. Run on a **6,000 GPU-hour** NCSA DeltaAI allocation (PHY250286).
- Outstanding Reviewer, ICML 2026 Mechanistic Interpretability Workshop. Invited talks at Harvard CfA (AstroAI, Jan 2026) and NeurIPS ML4PS (oral, Dec 2025).

Space Physics Lab, Boston University

Boston, MA

Graduate Research Assistant

Oct 2022 – May 2024

- Reconstructed corrupted sensor data with **generative inpainting** and **SwinIR super-resolution**, improving signal-to-noise for downstream models by over **60%**; M.S. thesis on deep learning over noisy all-sky imagery.
- Architected distributed ingestion (Kafka, Dask, S3) cutting processing of 3 TB+ datasets from **>24 h** to **<3 h**; built a forecasting service at **<80 ms p90** and 25 predictions/sec via async batching.

SELECTED PUBLICATIONS

1. **K. Duraphe**, et al. *Shared Geometry Is Not Shared Physics: A Layerwise Test of the Platonic Representation Hypothesis in Astronomy*. **NeurIPS Interpretability for Discovery Workshop**, 2026.

2. **K. Duraphe**, A. Kumar, S. Sourav, M. J. Smith. *What AstroPT knows about galaxies, and what that can teach us about LLMs*. **Sci-FM Workshop, COLM**, 2026.

3. **K. Duraphe**, M. J. Smith, J. F. Wu, S. Sourav (co-first). *The Platonic Universe: Do Foundation Models See the Same Sky?* **NeurIPS ML4PS Workshop**, 2025 — **Spotlight (top 1%)**. Expanded version submitted to ICLR 2027.

4. **K. Duraphe**, G. Bhatta, et al. *State-Dependent X-ray Variability in Cygnus X-1: A 12-Year NuSTAR Timing Study*. **The Astrophysical Journal**, 2026.

OPEN SOURCE

platonic-universe	Foundation-model representational-alignment evaluation framework for astronomy.
rinexpy	Fast RINEX/GNSS reading and correction library; superset of georinex .
jobber	Declarative job submission to cloud GPU providers.
ArkOS (MIT SIPB)	DevOps and documentation for an open-source local-LLM agent platform.

TECHNICAL SKILLS

Languages:	Python (primary), C, C++, CUDA, TypeScript, SQL (Postgres)
ML systems:	PyTorch, FSDP, distributed training, custom CUDA kernels, GPU profiling, quantization (GPTQ, 8-bit), distillation, edge inference and model compilation, RL post-training, VQ-VAE, GNNs, VLMs
Infrastructure:	ClearML, MLflow, Slurm, Kubernetes, Docker, Terraform, Azure, AWS, GCP, Kafka, Dask, Datadog, GitHub Actions
Methods:	Evaluation design and measurement, representational analysis (CKA, mutual k -NN), Bayesian inference (MCMC), uncertainty quantification, time-series and spectral analysis, inverse problems

EDUCATION

Boston University	Boston, MA
Master of Science with Thesis in Electrical and Computer Engineering	Sep 2022 – May 2024 · GPA: 3.8/4
College of Engineering Pune	Pune, India
Bachelor of Technology in Electrical Engineering, Minor in CS	Aug 2018 – June 2022 · GPA: 3.83/4